

Practice Worksheet 4

Section 1: Multiple Choice Questions

(No calculator is allowed for these questions.)

1. How many solutions does the equation $2 \cos^2 x - \cos x - 3 = 0$ have on the interval $[0, 2\pi)$?

(A) 0

(B) 1

(C) 2

(D) 3

2. What is the sum of all solutions to the equation $\sin(2x) = \cos x$ on the interval $[0, 2\pi)$?

(A) 2π

(B) $\frac{5\pi}{2}$

(C) 3π

(D) 4π

3. What are the equations of the vertical asymptotes of the graph of $y = 3 \csc\left(2x - \frac{\pi}{3}\right)$ on the interval $[0, \pi]$?

(A) $x = \frac{\pi}{6}$ and $x = \frac{2\pi}{3}$

(B) $x = \frac{\pi}{3}$ and $x = \frac{5\pi}{6}$

(C) $x = \frac{\pi}{6}$ only

(D) $x = 0$, $x = \frac{\pi}{2}$, and $x = \pi$

4. Which of the following intervals represents the solution set to the inequality $\tan^2 x < 1$ on the interval $[0, \pi]$?

(A) $\left(0, \frac{\pi}{4}\right)$

(B) $\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$

(C) $\left[0, \frac{\pi}{4}\right) \cup \left(\frac{3\pi}{4}, \pi\right]$

(D) $\left(0, \frac{\pi}{4}\right) \cup \left(\frac{3\pi}{4}, \pi\right)$

5. Which of the following is equivalent to the expression $\sin\left(x + \frac{\pi}{2}\right) - \cos(x - \pi)$ for all real numbers x ?

- (A) 0
- (B) $2 \sin x$
- (C) $2 \cos x$
- (D) $-2 \cos x$

6. Given that $\tan \theta = -\frac{4}{3}$ and $\sin \theta > 0$, what is the exact value of $\sec \theta - \csc \theta$?

- (A) $-\frac{35}{12}$
- (B) $-\frac{5}{12}$
- (C) $\frac{7}{12}$
- (D) $\frac{35}{12}$

7. Which of the following expressions is equivalent to $\frac{1+\cos(2x)}{\sin(2x)}$ for all x where the expression is defined?

- (A) $\tan x$
- (B) $\cot x$
- (C) $\sin x$
- (D) $\sec x$

8. How many solutions does the equation $2 \sin x \tan x = 3$ have on the interval $[0, 2\pi)$?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

9. For how many values of x on the interval $[0, 2\pi)$ is the function $f(x) = \frac{\sec x}{\tan x - 1}$ undefined?

- (A) 2
- (B) 3
- (C) 4

(D) 5

10. If $\cos A = \frac{1}{3}$, what is the exact value of $\cos(4A)$?

(A) $-\frac{7}{9}$ (B) $\frac{1}{81}$ (C) $\frac{17}{81}$ (D) $\frac{31}{81}$

11. How many solutions does the equation $3 \csc\left(\frac{x}{2}\right) - 2 = 0$ have on the interval $[0, 4\pi)$?

(A) 0

(B) 1

(C) 2

(D) 4

12. Which of the following expressions is equivalent to $\frac{1}{\sec x - \tan x}$ for all x where the expression is defined?

(A) $\sec x + \tan x$ (B) $\cos x + \cot x$ (C) $\frac{1}{\cos x + \sin x}$ (D) $\sec^2 x + \tan^2 x$

Section 2: Free Response Questions

(Show all your work clearly.)

FRQ 1

Let $f(x) = \cos(2x) + 3 \sin x - 2$.

(a) Find all solutions to the equation $f(x) = 0$ on the interval $[0, 2\pi)$.

(b) Determine the intervals on $[0, 2\pi)$ for which $f(x) > 0$. Justify your reasoning.

FRQ 2

Let angle α be such that $\sin \alpha = -\frac{3}{5}$ and $\pi < \alpha < \frac{3\pi}{2}$. Let angle β be such that $\sec \beta = \frac{13}{5}$ and $0 < \beta < \frac{\pi}{2}$.

- (a) Find the exact value of $\sin(\alpha + \beta)$ and $\cos(\alpha - \beta)$.
- (b) Find the exact value of $\tan(2\alpha)$.
- (c) Determine whether the terminal side of angle 2α is located in Quadrant I, II, III, or IV. Justify your reasoning.

FRQ 3

Consider the function $g(x) = \frac{\cot x - \tan x}{\cot x + \tan x}$.

- (a) Use trigonometric identities to prove that $g(x) = \cos(2x)$ for all x in the domain of g .
- (b) Solve the equation $g(x) = -\frac{1}{2}$ on the interval $[0, \pi)$.
- (c) Evaluate $g\left(\frac{\pi}{8}\right)$. Give your answer in simplest radical form.

FRQ 4

Let $h(x) = \sin x \cos x$.

- (a) Rewrite $h(x)$ in the form $A \sin(Bx)$, where A and B are constants.
- (b) Find all values of x on the interval $[0, \pi]$ such that $h(x) = \frac{\sqrt{3}}{4}$.
- (c) Given that $\sin\left(\frac{\pi}{2} - x\right) = \frac{1}{3}$ and x is in Quadrant I, determine the exact value of $h(x)$.